SRI: N06LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 6$. The Hubble-flow parameters are denoted by ϵ_j and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

1 Spectra

1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*}} \left[\sum_{d=0}^{6} F_{s,d} + O(\mathrm{SR}^7)\right].$$

Relative order 0.

$$F_{s,0} = 1.0$$

Relative order 1.

$$F_{s,1} = 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1$$

Relative order 2.

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 0.14504832807052\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 3.

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.959600275274\epsilon_1\epsilon_2^2 \\ & + 0.030669313835717\epsilon_2\epsilon_3^2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\ & + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \end{aligned}$$

Relative order 4.

$$\begin{aligned} F_{s,4} = {} & 0.083467005785127\epsilon_2^4 + 2.9412556561294\epsilon_1^4 \\ & + 0.055787239966231\epsilon_2\epsilon_3^3 + 8.1576664436908\epsilon_1^2\epsilon_2^2 \\ & + 9.1655563981061\epsilon_1^3\epsilon_2 + 2.2830486720555\epsilon_1\epsilon_2^3 \\ & - 0.20996470057763\epsilon_2^3\epsilon_3 - 0.058660421836305\epsilon_2^2\epsilon_3^2 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4^2 + 0.16736171989869\epsilon_2\epsilon_3^2\epsilon_4 \\ & + 0.38664546255166\epsilon_1^2\epsilon_2\epsilon_3 - 0.016062214045699\epsilon_2^2\epsilon_3\epsilon_4 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3^2 - 0.91389525644738\epsilon_1\epsilon_2^2\epsilon_3 \\ & + 0.055787239966231\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\ & - 0.079171041433698\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{aligned}$$

Relative order 5.

$$\begin{aligned} F_{s,5} = {} & 5.0146640695735\epsilon_1^5 + 0.023060510027566\epsilon_2^5 \\ & + 27.144156780677\epsilon_1^3\epsilon_2^2 + 0.018082530239889\epsilon_2\epsilon_3^4 \\ & + 18.923628528471\epsilon_1^4\epsilon_2 + 2.4064328576811\epsilon_1\epsilon_2^4 \\ & + 0.15472627942479\epsilon_2^3\epsilon_3^2 + 14.228251540965\epsilon_1^2\epsilon_2^3 \\ & - 0.13890612714825\epsilon_2^4\epsilon_3 - 0.10759865658691\epsilon_2^2\epsilon_3^3 \end{aligned}$$

$$
\begin{aligned}
&+ 2.0842227586428\epsilon_1^2\epsilon_2\epsilon_3^2 + 0.59532818615497\epsilon_1\epsilon_2^2\epsilon_3^2 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3^3 + 2.9656408144517\epsilon_1^3\epsilon_2\epsilon_3 \\
&+ 0.12657771167922\epsilon_2\epsilon_3^2\epsilon_4^2 + 0.06391567355745\epsilon_2^3\epsilon_3\epsilon_4 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4^3 + 0.78872771949954\epsilon_1\epsilon_2^3\epsilon_3 \\
&+ 3.8758981992336\epsilon_1^2\epsilon_2^2\epsilon_3 + 0.0049502975692885\epsilon_2^2\epsilon_3\epsilon_4^2 \\
&+ 0.10849518143933\epsilon_2\epsilon_3^3\epsilon_4 - 0.097698061448331\epsilon_2^2\epsilon_3^2\epsilon_4 \\
&+ 2.0842227586428\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4\epsilon_5^2 \\
&+ 0.072330120959556\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5 \\
&+ 1.1742909471328\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4 \\
&+ 0.0049502975692885\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.054247590719667\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5 \\
&+ 0.2397726251184\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4 \\
&+ 0.079924208372801\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.018082530239889\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6
\end{aligned}
$$

Relative order 6.

$$
\begin{aligned}
F_{s,6} = {} & 8.3167926130457\epsilon_1^6 + 0.0076684970179054\epsilon_2^6 \\
&+ 37.072173733221\epsilon_1^5\epsilon_2 + 72.521097458045\epsilon_1^3\epsilon_2^3 \\
&+ 0.097604164250968\epsilon_2^4\epsilon_3^2 + 26.371954409687\epsilon_1^2\epsilon_2^4 \\
&+ 75.454092624244\epsilon_1^4\epsilon_2^2 + 2.4764082020822\epsilon_1\epsilon_2^5 \\
&+ 0.1036516324715\epsilon_2^3\epsilon_3^3 + 0.010323841911612\epsilon_2\epsilon_3^5 \\
&- 0.065946975163844\epsilon_2^5\epsilon_3 - 0.15390619513867\epsilon_2^2\epsilon_3^4 \\
&+ 0.0048539557736345\epsilon_1\epsilon_2\epsilon_3^4 + 5.3721189690976\epsilon_1\epsilon_2^4\epsilon_3 \\
&+ 0.10323841911612\epsilon_2\epsilon_3^4\epsilon_4 + 0.029930407425899\epsilon_2^4\epsilon_3\epsilon_4 \\
&+ 2.1503772818914\epsilon_1^2\epsilon_2\epsilon_3^3 + 0.095195191276171\epsilon_1\epsilon_2^2\epsilon_3^3 \\
&+ 18.232367303381\epsilon_1^2\epsilon_2^2\epsilon_3^2 + 0.15485762867417\epsilon_2\epsilon_3^2\epsilon_4^3 \\
&+ 5.6575079379471\epsilon_1\epsilon_2^3\epsilon_3^2 + 0.25809604779029\epsilon_2\epsilon_3^3\epsilon_4^2 \\
&+ 10.057811129131\epsilon_1^4\epsilon_2\epsilon_3 + 40.749413397007\epsilon_1^3\epsilon_2^2\epsilon_3 \\
&+ 9.5864240626752\epsilon_1^3\epsilon_2\epsilon_3^2 + 0.020302689804444\epsilon_2^3\epsilon_3\epsilon_4^2 \\
&+ 0.010323841911612\epsilon_2\epsilon_3\epsilon_4^4 + 0.10071828009788\epsilon_2^3\epsilon_3^2\epsilon_4 \\
&+ 27.47321752324\epsilon_1^2\epsilon_2^3\epsilon_3 - 0.014245226035509\epsilon_2^2\epsilon_3\epsilon_4^3 \\
&- 0.44525600202513\epsilon_2^2\epsilon_3^3\epsilon_4 - 0.23937755135172\epsilon_2^2\epsilon_3^2\epsilon_4^2 \\
&+ 0.0048539557736345\epsilon_1\epsilon_2\epsilon_3\epsilon_4^3 \\
&+ 0.051619209558058\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5^2 \\
&+ 0.10323841911612\epsilon_2\epsilon_3^3\epsilon_4\epsilon_5 \\
&+ 0.029123734641807\epsilon_1\epsilon_2\epsilon_3^3\epsilon_4 \\
&+ 2.1503772818914\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4^2 \\
&+ 0.033977690415441\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4^2 \\
&+ 1.1312985085359\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4^2 \\
&+ 6.4511318456741\epsilon_1^2\epsilon_2\epsilon_3^2\epsilon_4 \\
&+ 0.25809604779029\epsilon_2\epsilon_3^2\epsilon_4^2\epsilon_5
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.072266893381281\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5^2 \\
&+ 5.1348555860916\epsilon_1\epsilon_2^3\epsilon_3\epsilon_4 \\
&+ 0.061943051469669\epsilon_2\epsilon_3\epsilon_4^3\epsilon_5 \\
&+ 2.357792208348\epsilon_1\epsilon_2^2\epsilon_3^2\epsilon_4 \\
&+ 9.5864240626752\epsilon_1^3\epsilon_2\epsilon_3\epsilon_4 \\
&+ 0.020302689804444\epsilon_2^3\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.010323841911612\epsilon_2\epsilon_3\epsilon_4\epsilon_5^3 \\
&+ 15.388135466879\epsilon_1^2\epsilon_2^2\epsilon_3\epsilon_4 \\
&- 0.014245226035509\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5^2 \\
&- 0.13744361174779\epsilon_2^2\epsilon_3^2\epsilon_4\epsilon_5 \\
&- 0.042735678106526\epsilon_2^2\epsilon_3\epsilon_4^2\epsilon_5 \\
&+ 0.0048539557736345\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5^2 \\
&+ 0.019415823094538\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5 \\
&+ 0.051619209558058\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5\epsilon_6 \\
&+ 0.014561867320903\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5 \\
&+ 2.1503772818914\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 1.1312985085359\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5 \\
&+ 0.030971525734835\epsilon_2\epsilon_3\epsilon_4\epsilon_5^2\epsilon_6 \\
&+ 0.010323841911612\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6^2 \\
&+ 0.041295367646446\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5\epsilon_6 \\
&- 0.014245226035509\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5\epsilon_6 \\
&+ 0.0048539557736345\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6 \\
&+ 0.010323841911612\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6\epsilon_7
\end{aligned}
$$

1.2 Tensor spectrum

$$
P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2}\left[\sum_{d=0}^{6} F_{t,d} + O(\mathrm{SR}^7)\right].
$$

Relative order 0.

$$
F_{t,0} = 1.0
$$

Relative order 1.

$$
F_{t,1} = -0.54072569092296\epsilon_1
$$

Relative order 2.

$$
F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2
$$

Relative order 3.

$$
\begin{aligned}
F_{t,3} = {} & 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2 \\
& - 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3
\end{aligned}
$$

Relative order 4.

$$
\begin{aligned}
F_{t,4} = {} & 2.9412556561294\epsilon_1^4 + 4.0859368676311\epsilon_1^3\epsilon_2 \\
& + 1.8087367742501\epsilon_1^2\epsilon_2^2 - 0.077715927178017\epsilon_1\epsilon_2^3
\end{aligned}
$$

$$+ 1.696905383796\epsilon_1^2\epsilon_2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3^2$$
$$- 0.23314778153405\epsilon_1\epsilon_2^2\epsilon_3 - 0.077715927178017\epsilon_1\epsilon_2\epsilon_3\epsilon_4$$

Relative order 5.

$$\begin{aligned}
F_{t,5} &= 5.0146640695735\epsilon_1^5 + 10.182162879741\epsilon_1^3\epsilon_2^2 \\
&+ 1.9912795134148\epsilon_1^2\epsilon_2^3 + 9.3282240106668\epsilon_1^4\epsilon_2 \\
&- 0.041550866698239\epsilon_1\epsilon_2^4 + 5.715558726935\epsilon_1^2\epsilon_2^2\epsilon_3 \\
&+ 6.2270085551926\epsilon_1^3\epsilon_2\epsilon_3 + 1.8621396067601\epsilon_1^2\epsilon_2\epsilon_3^2 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3^3 - 0.29085606688767\epsilon_1\epsilon_2^2\epsilon_3^2 \\
&- 0.24930520018943\epsilon_1\epsilon_2^3\epsilon_3 + 1.8621396067601\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2 \\
&- 0.16620346679296\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4 \\
&- 0.12465260009472\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4 \\
&- 0.041550866698239\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5
\end{aligned}$$

Relative order 6.

$$\begin{aligned}
F_{t,6} &= 8.3167926130457\epsilon_1^6 + 35.569920863237\epsilon_1^4\epsilon_2^2 \\
&+ 25.766288824979\epsilon_1^3\epsilon_2^3 + 1.9852568085143\epsilon_1^2\epsilon_2^4 \\
&+ 19.643791998445\epsilon_1^5\epsilon_2 - 0.020903182875016\epsilon_1\epsilon_2^5 \\
&+ 1.8721030173592\epsilon_1^2\epsilon_2\epsilon_3^3 + 17.436687309176\epsilon_1^4\epsilon_2\epsilon_3 \\
&+ 42.955365634287\epsilon_1^3\epsilon_2^2\epsilon_3 + 11.663061184146\epsilon_1^2\epsilon_2^3\epsilon_3 \\
&+ 8.6456966910171\epsilon_1^3\epsilon_2\epsilon_3^2 + 13.217874912669\epsilon_1^2\epsilon_2^2\epsilon_3^2 \\
&- 0.31354774312524\epsilon_1\epsilon_2^2\epsilon_3^3 - 0.52257957187539\epsilon_1\epsilon_2^3\epsilon_3^2 \\
&- 0.020903182875016\epsilon_1\epsilon_2\epsilon_3^4 - 0.20903182875016\epsilon_1\epsilon_2^4\epsilon_3 \\
&+ 1.8721030173592\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4^2 \\
&+ 5.6163090520775\epsilon_1^2\epsilon_2\epsilon_3^2\epsilon_4 \\
&+ 7.692547567117\epsilon_1^2\epsilon_2^2\epsilon_3\epsilon_4 \\
&+ 8.6456966910171\epsilon_1^3\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.14632228012511\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4^2 \\
&- 0.12541909725009\epsilon_1\epsilon_2\epsilon_3^3\epsilon_4 \\
&- 0.52257957187539\epsilon_1\epsilon_2^2\epsilon_3^2\epsilon_4 \\
&- 0.020903182875016\epsilon_1\epsilon_2\epsilon_3\epsilon_4^3 \\
&- 0.10451591437508\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4^2 \\
&- 0.20903182875016\epsilon_1\epsilon_2^3\epsilon_3\epsilon_4 \\
&+ 1.8721030173592\epsilon_1^2\epsilon_2\epsilon_3\epsilon_4\epsilon_5 \\
&- 0.062709548625047\epsilon_1\epsilon_2\epsilon_3\epsilon_4^2\epsilon_5 \\
&- 0.020903182875016\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5^2 \\
&- 0.083612731500063\epsilon_1\epsilon_2\epsilon_3^2\epsilon_4\epsilon_5 \\
&- 0.10451591437508\epsilon_1\epsilon_2^2\epsilon_3\epsilon_4\epsilon_5 \\
&- 0.020903182875016\epsilon_1\epsilon_2\epsilon_3\epsilon_4\epsilon_5\epsilon_6
\end{aligned}$$

2 Consistency relations

2.1 $n = 0$

For $p = 6$, the absolute cutoff is $W = 7$:

$$\alpha_0^t = \sum_{d=1}^{7} R_{0,d} + O(\mathrm{SR}^8).$$

Absolute order 1.

$$R_{0,1} = -0.125r$$

Absolute order 2.

$$R_{0,2} = 0.0078125r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned} R_{0,3} = &-0.0030712246746565r^3 + 0.125\alpha_1^s r \\ &- 0.10280837917801\left(\alpha_0^s\right)^2 r - 0.041327094794504\alpha_0^s r^2 \end{aligned}$$

Absolute order 4.

$$\begin{aligned} R_{0,4} = &\ 0.0014926374043055r^4 + 0.125\alpha_2^s r \\ &+ 0.065244100954277\left(\alpha_0^s\right)^3 r + 0.024554893470954\alpha_0^s r^3 \\ &+ 0.10320649151137\left(\alpha_0^s\right)^2 r^2 - 0.060109233906372\alpha_1^s r^2 \\ &- 0.23087387125098\alpha_0^s \alpha_1^s r \end{aligned}$$

Absolute order 5.

$$\begin{aligned} R_{0,5} = &-0.00093280100983415r^5 + 0.125\alpha_3^s r \\ &+ 0.04315951309091\alpha_1^s r^3 - 0.038536574146577\left(\alpha_0^s\right)^4 r \\ &- 0.018206214478227\alpha_0^s r^4 - 0.077020534433109\alpha_2^s r^2 \\ &- 0.14103704832205\left(\alpha_1^s\right)^2 r - 0.18695785481671\left(\alpha_0^s\right)^3 r^2 \\ &- 0.10777687087114\left(\alpha_0^s\right)^2 r^3 + 0.33322659757276\alpha_0^s \alpha_1^s r^2 \\ &+ 0.25691700713838\left(\alpha_0^s\right)^2 \alpha_1^s r - 0.24116427546487\alpha_0^s \alpha_2^s r \end{aligned}$$

Absolute order 6.

$$\begin{aligned} R_{0,6} = &\ 0.00070298503785145r^6 + 0.125\alpha_4^s r \\ &+ 0.020902477635746\left(\alpha_0^s\right)^5 r + 0.015994052232973\alpha_0^s r^5 \\ &+ 0.065362321729807\alpha_2^s r^3 + 0.28417656846613\left(\alpha_0^s\right)^4 r^2 \\ &+ 0.30605471800346\left(\alpha_1^s\right)^2 r^2 + 0.34969699644176\left(\alpha_0^s\right)^3 r^3 \\ &+ 0.12218812205202\left(\alpha_0^s\right)^2 r^4 - 0.093222530607224\alpha_3^s r^2 \\ &- 0.037217907105313\alpha_1^s r^4 + 0.4688109212146\alpha_0^s \alpha_2^s r^2 \\ &+ 0.37293859690816\alpha_0^s\left(\alpha_1^s\right)^2 r + 0.29046664127673\left(\alpha_0^s\right)^2 \alpha_2^s r \\ &- 0.22801627965519\left(\alpha_0^s\right)^3 \alpha_1^s r - 1.015918735308\left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\ &- 0.31399132156735\alpha_1^s \alpha_2^s r - 0.42145582743858\alpha_0^s \alpha_1^s r^3 \\ &- 0.24578024485779\alpha_0^s \alpha_3^s r \end{aligned}$$

Absolute order 7.

$$\begin{aligned}R_{0,7} = &-0.00061515419372413r^7 + 0.125\alpha_5^s r \\ &+ 0.03707739098979\alpha_1^s r^5 + 0.19980707266399\left(\alpha_1^s\right)^3 r \\ &+ 0.091144689305009\alpha_3^s r^3 - 0.016058051152622\alpha_0^s r^6 \\ &- 0.011049369381434\left(\alpha_0^s\right)^6 r - 0.18310571021651\left(\alpha_2^s\right)^2 r \\ &- 0.94736847967917\left(\alpha_0^s\right)^4 r^3 - 0.064533532306596\alpha_2^s r^4 \\ &- 0.60450331655168\left(\alpha_0^s\right)^3 r^4 - 0.15064645355686\left(\alpha_0^s\right)^2 r^5 \\ &- 0.38746735293638\left(\alpha_0^s\right)^5 r^2 - 0.10911851595073\alpha_4^s r^2 \\ &- 0.46539756860253\left(\alpha_1^s\right)^2 r^3 + 0.55277396843725\alpha_0^s\alpha_1^s r^4 \\ &+ 2.2882998446743\left(\alpha_0^s\right)^2\alpha_1^s r^3 + 0.61514830647938\alpha_0^s\alpha_3^s r^2 \\ &+ 0.17617169912302\left(\alpha_0^s\right)^4\alpha_1^s r + 0.97696202212274\alpha_1^s\alpha_2^s r^2 \\ &+ 2.3093282152193\left(\alpha_0^s\right)^3\alpha_1^s r^2 + 0.30878621388554\left(\alpha_0^s\right)^2\alpha_3^s r \\ &- 1.5812089463636\left(\alpha_0^s\right)^2\alpha_2^s r^2 \\ &- 0.56258489487217\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r - 0.33131930046622\alpha_1^s\alpha_3^s r \\ &- 0.24794812760585\alpha_0^s\alpha_4^s r - 2.0996792408229\alpha_0^s\left(\alpha_1^s\right)^2 r^2 \\ &- 0.28144573547165\left(\alpha_0^s\right)^3\alpha_2^s r - 0.7030684429864\alpha_0^s\alpha_2^s r^3 \\ &+ 0.90387358971276\alpha_0^s\alpha_1^s\alpha_2^s r\end{aligned}$$

2.2 $n = 1$

For $p = 6$, the absolute cutoff is $W = 8$:

$$\alpha_1^t = \sum_{d=2}^{8} R_{1,d} + O(\mathrm{SR}^9).$$

Absolute order 2.

$$R_{1,2} = 0.015625r^2 + 0.125\alpha_0^s r$$

Absolute order 3.

$$\begin{aligned}R_{1,3} = &-0.0029296875r^3 + 0.125\alpha_1^s r \\ &- 0.125\left(\alpha_0^s\right)^2 r - 0.046875\alpha_0^s r^2\end{aligned}$$

Absolute order 4.

$$\begin{aligned}R_{1,4} = &0.0016576826498282r^4 + 0.125\alpha_2^s r \\ &+ 0.12398128438351\left(\alpha_0^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^3 r \\ &+ 0.027641022071908\alpha_0^s r^3 - 0.072577094794504\alpha_1^s r^2 \\ &- 0.33061675835603\alpha_0^s\alpha_1^s r\end{aligned}$$

Absolute order 5.

$$\begin{aligned}R_{1,5} = &-0.0010528680915447r^5 + 0.125\alpha_3^s r \\ &+ 0.050025543685844\alpha_1^s r^3 - 0.091359233906372\alpha_2^s r^2 \\ &- 0.065244100954277\left(\alpha_0^s\right)^4 r - 0.13027434666455\left(\alpha_0^s\right)^2 r^3 \\ &- 0.021075080402355\alpha_0^s r^4 - 0.23087387125098\left(\alpha_1^s\right)^2 r \\ &- 0.24842610305581\left(\alpha_0^s\right)^3 r^2 + 0.42660617411381\left(\alpha_0^s\right)^2\alpha_1^s r \\ &+ 0.41559991864823\alpha_0^s\alpha_1^s r^2 - 0.35587387125098\alpha_0^s\alpha_2^s r\end{aligned}$$

Absolute order 6.

$$
\begin{aligned}
R_{1,6} = {} & 0.00079786539870909 r^6 + 0.125 \alpha_4^s r \\
& + 0.038536574146577 \left(\alpha_0^s\right)^5 r + 0.41043044123784 \left(\alpha_0^s\right)^4 r^2 \\
& + 0.14689479099556 \left(\alpha_0^s\right)^2 r^4 + 0.018569586968286 \alpha_0^s r^5 \\
& + 0.074971901003326 \alpha_2^s r^3 + 0.38411085965327 \left(\alpha_1^s\right)^2 r^2 \\
& + 0.44641771599844 \left(\alpha_0^s\right)^3 r^3 - 0.043199996920584 \alpha_1^s r^4 \\
& - 0.10827053443311 \alpha_3^s r^2 + 0.5788087353052 \alpha_0^s \alpha_2^s r^2 \\
& + 0.65487106259881 \alpha_0^s \left(\alpha_1^s\right)^2 r + 0.49808128260325 \left(\alpha_0^s\right)^2 \alpha_2^s r \\
& - 0.36616427546487 \alpha_0^s \alpha_3^s r - 0.52323837210896 \alpha_1^s \alpha_2^s r \\
& - 0.51344198472507 \alpha_0^s \alpha_1^s r^3 - 1.3749765739875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\
& - 0.41106330372469 \left(\alpha_0^s\right)^3 \alpha_1^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{1,7} = {} & -0.00069821403713025 r^7 + 0.125 \alpha_5^s r \\
& + 0.37293859690816 \left(\alpha_1^s\right)^3 r + 0.042797105843338 \alpha_1^s r^5 \\
& + 0.10325045820752 \alpha_3^s r^3 - 0.75129206502539 \left(\alpha_0^s\right)^3 r^4 \\
& - 0.12447253060722 \alpha_4^s r^2 - 0.17935279734314 \left(\alpha_0^s\right)^2 r^5 \\
& - 0.020902477635746 \left(\alpha_0^s\right)^6 r - 0.56207286522481 \left(\alpha_1^s\right)^2 r^3 \\
& - 0.073841572008579 \alpha_2^s r^4 - 0.018589066236159 \alpha_0^s r^6 \\
& - 0.59662818041792 \left(\alpha_0^s\right)^5 r^2 - 0.31399132156735 \left(\alpha_2^s\right)^2 r \\
& - 1.2665901227865 \left(\alpha_0^s\right)^4 r^3 + 1.1906681876134 \alpha_1^s \alpha_2^s r^2 \\
& + 0.53624688613452 \left(\alpha_0^s\right)^2 \alpha_3^s r + 0.7479510436435 \alpha_0^s \alpha_3^s r^2 \\
& + 0.66101914285082 \alpha_0^s \alpha_1^s r^4 + 0.33252866783392 \left(\alpha_0^s\right)^4 \alpha_1^s r \\
& + 2.9112751359658 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 + 3.3535596284132 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \\
& - 2.1121504017171 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 - 0.84337953675004 \alpha_0^s \alpha_2^s r^3 \\
& - 0.51848292093192 \left(\alpha_0^s\right)^3 \alpha_2^s r \\
& - 1.0569874358737 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r - 0.55977156642514 \alpha_1^s \alpha_3^s r \\
& - 2.8301592857432 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 - 0.37078024485779 \alpha_0^s \alpha_4^s r \\
& + 1.6408017979371 \alpha_0^s \alpha_1^s \alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{1,8} = {} & 0.00069092135609635 r^8 + 0.125 \alpha_6^s r \\
& + 0.020764478937666 \alpha_0^s r^7 + 0.011049369381434 \left(\alpha_0^s\right)^7 r \\
& + 1.2687327364727 \left(\alpha_0^s\right)^3 r^5 + 3.1771029716803 \left(\alpha_0^s\right)^5 r^3 \\
& + 3.1788385090986 \left(\alpha_0^s\right)^4 r^4 + 0.82557558507459 \left(\alpha_1^s\right)^2 r^4 \\
& + 1.0383634496769 \left(\alpha_2^s\right)^2 r^2 + 0.13499382028653 \alpha_4^s r^3 \\
& + 0.082079699580705 \alpha_2^s r^5 + 0.79510322579056 \left(\alpha_0^s\right)^6 r^2 \\
& + 0.23686528225709 \left(\alpha_0^s\right)^2 r^6 - 0.1145546933779 \alpha_3^s r^4 \\
& - 2.1848902710694 \left(\alpha_1^s\right)^3 r^2 - 0.14036851595073 \alpha_5^s r^2 \\
& - 0.047643927662444 \alpha_1^s r^6 + 1.2377460864495 \alpha_0^s \alpha_2^s r^4 \\
& + 7.0677622611573 \alpha_0^s \left(\alpha_1^s\right)^2 r^3 + 5.2654277957421 \left(\alpha_0^s\right)^2 \alpha_2^s r^3 \\
& + 0.55673434149138 \left(\alpha_0^s\right)^2 \alpha_4^s r
\end{aligned}
$$

$$
\begin{aligned}
&+ 11.507755577144\,(\alpha_0^s)^2\,(\alpha_1^s)^2\,r^2 \\
&+ 1.5032948077047\,(\alpha_1^s)^2\,\alpha_2^s r + 0.45761743459468\,(\alpha_0^s)^4\,\alpha_2^s r \\
&+ 1.0869792999293\alpha_0^s\,(\alpha_2^s)^2\,r \\
&+ 1.2672716913643\,(\alpha_0^s)^3\,(\alpha_1^s)^2\,r \\
&+ 5.6806226439244\,(\alpha_0^s)^3\,\alpha_2^s r^2 + 1.7061901537187\alpha_1^s\alpha_3^s r^2 \\
&+ 0.9266223702823\alpha_0^s\alpha_4^s r^2 - 1.3249768624083\alpha_0^s\,(\alpha_1^s)^3\,r \\
&- 0.57926742807207\alpha_1^s\alpha_4^s r - 0.59023194935719\,(\alpha_0^s)^3\,\alpha_3^s r \\
&- 1.2633997098457\alpha_0^s\alpha_3^s r^3 - 12.194671111486\,(\alpha_0^s)^3\,\alpha_1^s r^3 \\
&- 2.059007173214\alpha_1^s\alpha_2^s r^3 - 0.69753072089924\alpha_2^s\alpha_3^s r \\
&- 5.5886005276836\,(\alpha_0^s)^2\,\alpha_1^s r^4 - 2.9758492688027\,(\alpha_0^s)^2\,\alpha_3^s r^2 \\
&- 0.37294812760585\alpha_0^s\alpha_5^s r - 6.7496270918532\,(\alpha_0^s)^4\,\alpha_1^s r^2 \\
&- 0.90640507582801\alpha_0^s\alpha_1^s r^5 - 0.24246791541163\,(\alpha_0^s)^5\,\alpha_1^s r \\
&+ 1.8527653179501\alpha_0^s\alpha_1^s\alpha_3^s r - 9.7381761831174\alpha_0^s\alpha_1^s\alpha_2^s r^2 \\
&- 2.8733805858721\,(\alpha_0^s)^2\,\alpha_1^s\alpha_2^s r
\end{aligned}
$$

2.3 $n = 2$

For $p = 6$, the absolute cutoff is $W = 9$:

$$\alpha_2^t = \sum_{d=3}^{9} R_{2,d} + O(\mathrm{SR}^{10}).$$

Absolute order 3.

$$
\begin{aligned}
R_{2,3} = &-0.00390625r^3 + 0.125\alpha_1^s r \\
&- 0.125\,(\alpha_0^s)^2\,r - 0.046875\alpha_0^s r^2
\end{aligned}
$$

Absolute order 4.

$$
\begin{aligned}
R_{2,4} = &\,0.0013427734375r^4 + 0.125\alpha_2^s r \\
&+ 0.125\,(\alpha_0^s)^3\,r + 0.125\,(\alpha_0^s)^2\,r^2 \\
&+ 0.025390625\alpha_0^s r^3 - 0.0625\alpha_1^s r^2 \\
&- 0.375\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{2,5} = &-0.00099348164677838r^5 + 0.125\alpha_3^s r \\
&+ 0.050668108270534\alpha_1^s r^3 - 0.13499234851023\,(\alpha_0^s)^2\,r^3 \\
&- 0.28928966356152\,(\alpha_0^s)^3\,r^2 - 0.020502543360439\alpha_0^s r^4 \\
&- 0.10280837917801\,(\alpha_0^s)^4\,r - 0.33061675835603\,(\alpha_1^s)^2\,r \\
&- 0.088202094794504\alpha_2^s r^2 + 0.63904189589007\,(\alpha_0^s)^2\,\alpha_1^s r \\
&+ 0.46569385315053\alpha_0^s\alpha_1^s r^2 - 0.45561675835603\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{2,6} = &\,0.0007834832445242r^6 + 0.125\alpha_4^s r \\
&+ 0.065244100954277\,(\alpha_0^s)^5\,r + 0.51447193930681\,(\alpha_0^s)^3\,r^3 \\
&+ 0.018883632267472\alpha_0^s r^5 + 0.077748164662437\alpha_2^s r^3 \\
&+ 0.5388653261447\,(\alpha_0^s)^4\,r^2 + 0.15691226448912\,(\alpha_0^s)^2\,r^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.4600841525546 \left(\alpha_1^s\right)^2 r^2 - 0.10698423390637 \alpha_3^s r^2 \\
&- 0.044329625847117 \alpha_1^s r^4 + 0.78248004536479 \left(\alpha_0^s\right)^2 \alpha_2^s r \\
&+ 1.0840862194786 \alpha_0^s \left(\alpha_1^s\right)^2 r + 0.67405262036735 \alpha_0^s \alpha_2^s r^2 \\
&- 0.56686562073618 \alpha_0^s \alpha_1^s r^3 - 0.68758257793092 \left(\alpha_0^s\right)^3 \alpha_1^s r \\
&- 0.81762161375293 \alpha_1^s \alpha_2^s r - 1.7284662943263 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \\
&- 0.48087387125098 \alpha_0^s \alpha_3^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{2,7} = &-0.00070216008727898 r^7 + 0.125 \alpha_5^s r \\
&+ 0.10692234711116 \alpha_3^s r^3 + 0.65487106259881 \left(\alpha_1^s\right)^3 r \\
&+ 0.044680319835584 \alpha_1^s r^5 - 0.038536574146577 \left(\alpha_0^s\right)^6 r \\
&- 0.076631375424485 \alpha_2^s r^4 - 0.85737561408011 \left(\alpha_0^s\right)^5 r^2 \\
&- 0.52323837210896 \left(\alpha_2^s\right)^2 r - 0.019279820845041 \alpha_0^s r^6 \\
&- 1.5778098507287 \left(\alpha_0^s\right)^4 r^3 - 0.12389553443311 \alpha_4^s r^2 \\
&- 0.64133873745452 \left(\alpha_1^s\right)^2 r^3 - 0.193795881651 \left(\alpha_0^s\right)^2 r^5 \\
&- 0.85944179815707 \left(\alpha_0^s\right)^3 r^4 + 1.4436852511254 \alpha_1^s \alpha_2^s r^2 \\
&+ 4.6123541905271 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 + 0.72830526168963 \alpha_0^s \alpha_1^s r^4 \\
&+ 0.87237033860452 \alpha_0^s \alpha_3^s r^2 + 0.60374617445758 \left(\alpha_0^s\right)^4 \alpha_1^s r \\
&+ 0.86424555806812 \left(\alpha_0^s\right)^2 \alpha_3^s r + 3.481878776513 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \\
&- 0.88940264757383 \alpha_1^s \alpha_3^s r - 0.94272853191881 \alpha_0^s \alpha_2^s r^3 \\
&- 1.8880609737729 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r - 0.49116427546487 \alpha_0^s \alpha_4^s r \\
&- 2.69796002066 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 - 0.90914458632794 \left(\alpha_0^s\right)^3 \alpha_2^s r \\
&- 3.7167089437539 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 + 2.8291430625131 \alpha_0^s \alpha_1^s \alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{2,8} = &0.00070456920068177 r^8 + 0.125 \alpha_6^s r \\
&+ 0.13925140335932 \alpha_4^s r^3 + 1.2215314043215 \left(\alpha_0^s\right)^6 r^2 \\
&+ 0.020902477635746 \left(\alpha_0^s\right)^7 r + 1.4381221889498 \left(\alpha_0^s\right)^3 r^5 \\
&+ 0.92219068354551 \left(\alpha_1^s\right)^2 r^4 + 3.8323296829683 \left(\alpha_0^s\right)^4 r^4 \\
&+ 0.25673070632081 \left(\alpha_0^s\right)^2 r^6 + 4.1955594060503 \left(\alpha_0^s\right)^5 r^3 \\
&+ 1.2455421028093 \left(\alpha_2^s\right)^2 r^2 + 0.085771147089679 \alpha_2^s r^5 \\
&+ 0.021751280469382 \alpha_0^s r^7 - 0.11864796091522 \alpha_3^s r^4 \\
&- 0.14009753060722 \alpha_5^s r^2 - 2.9232654753033 \left(\alpha_1^s\right)^3 r^2 \\
&- 0.050080440992255 \alpha_1^s r^6 + 2.03984067706 \alpha_1^s \alpha_3^s r^2 \\
&+ 2.3871021072094 \left(\alpha_0^s\right)^3 \left(\alpha_1^s\right)^2 r \\
&+ 1.9547931195045 \alpha_0^s \left(\alpha_2^s\right)^2 r + 0.90702713099232 \left(\alpha_0^s\right)^2 \alpha_4^s r \\
&+ 16.184733661745 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r^2 \\
&+ 8.5890660587772 \alpha_0^s \left(\alpha_1^s\right)^2 r^3 + 6.2953483110811 \left(\alpha_0^s\right)^2 \alpha_2^s r^3 \\
&+ 0.85101158876584 \left(\alpha_0^s\right)^4 \alpha_2^s r + 1.0744936354652 \alpha_0^s \alpha_4^s r^2 \\
&+ 1.3640509580442 \alpha_0^s \alpha_2^s r^4 + 2.7596175886616 \left(\alpha_1^s\right)^2 \alpha_2^s r \\
&+ 7.8289781977937 \left(\alpha_0^s\right)^3 \alpha_2^s r^2 - 0.99377356769359 \alpha_0^s \alpha_1^s r^5 \\
&- 2.4869134686556 \alpha_0^s \left(\alpha_1^s\right)^3 r - 1.1877542095598 \alpha_2^s \alpha_3^s r
\end{aligned}
$$

$$
\begin{aligned}
&- 1.4062650501208\alpha_0^s\alpha_3^s r^3 - 15.480967047284\left(\alpha_0^s\right)^3\alpha_1^s r^3 \\
&- 9.9459636515628\left(\alpha_0^s\right)^4\alpha_1^s r^2 - 3.7800524622085\left(\alpha_0^s\right)^2\alpha_3^s r^2 \\
&- 1.0547298070664\left(\alpha_0^s\right)^3\alpha_3^s r - 2.3372176459017\alpha_1^s\alpha_2^s r^3 \\
&- 6.4975922521103\left(\alpha_0^s\right)^2\alpha_1^s r^4 - 0.4579435336484\left(\alpha_0^s\right)^5\alpha_1^s r \\
&- 0.93055181128293\alpha_1^s\alpha_4^s r - 0.49578024485779\alpha_0^s\alpha_5^s r \\
&+ 3.2730671366313\alpha_0^s\alpha_1^s\alpha_3^s r - 5.3102254324804\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r \\
&- 12.720525783639\alpha_0^s\alpha_1^s\alpha_2^s r^2
\end{aligned}
$$

Absolute order 9.

$$
\begin{aligned}
R_{2,9} = &-0.00078325253131263 r^9 + 0.125\alpha_7^s r \\
&+ 0.061833969920015\alpha_1^s r^7 + 0.14712505634682\alpha_3^s r^5 \\
&+ 0.17496876177422\alpha_5^s r^3 + 7.7917767073786\left(\alpha_1^s\right)^3 r^3 \\
&- 8.7149027983697\left(\alpha_0^s\right)^4 r^5 - 1.3249768624083\left(\alpha_1^s\right)^4 r \\
&- 0.011049369381434\left(\alpha_0^s\right)^8 r - 0.026899939604252\alpha_0^s r^8 \\
&- 0.17201474879639\alpha_4^s r^4 - 1.3947772426699\left(\alpha_1^s\right)^2 r^5 \\
&- 0.36511047260991\left(\alpha_0^s\right)^2 r^7 - 14.90287048239\left(\alpha_0^s\right)^5 r^4 \\
&- 2.359240521558\left(\alpha_2^s\right)^2 r^3 - 2.4760833602147\left(\alpha_0^s\right)^3 r^6 \\
&- 0.15599351595073\alpha_6^s r^2 - 10.119576191149\left(\alpha_0^s\right)^6 r^3 \\
&- 0.69753072089924\left(\alpha_3^s\right)^2 r - 1.6103749714989\left(\alpha_0^s\right)^7 r^2 \\
&- 0.10609127961526\alpha_2^s r^6 + 3.3560601256548\left(\alpha_1^s\right)^2\alpha_3^s r \\
&+ 4.0935689153387\alpha_1^s\left(\alpha_2^s\right)^2 r + 1.0478493839519\left(\alpha_0^s\right)^4\alpha_3^s r \\
&+ 11.901103158999\left(\alpha_0^s\right)^3\alpha_3^s r^2 + 27.850915176513\alpha_0^s\left(\alpha_1^s\right)^3 r^2 \\
&+ 0.31981350108167\left(\alpha_0^s\right)^6\alpha_1^s r + 2.73647095251\alpha_1^s\alpha_4^s r^2 \\
&+ 12.044746060424\left(\alpha_0^s\right)^2\alpha_1^s r^5 + 56.260557283539\left(\alpha_0^s\right)^4\alpha_1^s r^3 \\
&+ 2.285169999955\alpha_0^s\alpha_3^s r^4 + 18.524012994936\left(\alpha_0^s\right)^5\alpha_1^s r^2 \\
&+ 10.193295128537\left(\alpha_0^s\right)^2\alpha_3^s r^3 \\
&+ 5.1267919365011\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^3 r + 1.2852279181345\alpha_0^s\alpha_5^s r^2 \\
&+ 42.176693101567\left(\alpha_0^s\right)^3\alpha_1^s r^4 + 1.4480298249474\alpha_0^s\alpha_1^s r^6 \\
&+ 0.92968246909723\left(\alpha_0^s\right)^2\alpha_5^s r + 3.7971280636148\alpha_1^s\alpha_2^s r^4 \\
&+ 3.9013583931848\alpha_2^s\alpha_3^s r^2 - 3.8280597428628\alpha_1^s\alpha_3^s r^3 \\
&- 5.0045028960089\left(\alpha_0^s\right)^2\alpha_4^s r^2 - 0.95221555567791\alpha_1^s\alpha_5^s r \\
&- 16.631901423945\left(\alpha_1^s\right)^2\alpha_2^s r^2 - 1.1469662908486\left(\alpha_0^s\right)^3\alpha_4^s r \\
&- 1.2767981489713\alpha_2^s\alpha_4^s r - 2.0713172476951\alpha_0^s\alpha_2^s r^5 \\
&- 17.984269601465\alpha_0^s\left(\alpha_1^s\right)^2 r^4 - 18.416565864046\left(\alpha_0^s\right)^4\alpha_2^s r^2 \\
&- 2.4796112684224\left(\alpha_0^s\right)^4\left(\alpha_1^s\right)^2 r \\
&- 0.70008535000631\left(\alpha_0^s\right)^5\alpha_2^s r - 12.087542392275\alpha_0^s\left(\alpha_2^s\right)^2 r^2 \\
&- 50.781699765006\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r^2 \\
&- 13.256871913634\left(\alpha_0^s\right)^2\alpha_2^s r^4 - 30.576918469482\left(\alpha_0^s\right)^3\alpha_2^s r^3 \\
&- 62.733882360978\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^3 - 1.9723244469215\alpha_0^s\alpha_4^s r^3 \\
&- 3.9603598858013\left(\alpha_0^s\right)^2\left(\alpha_2^s\right)^2 r - 0.49794812760585\alpha_0^s\alpha_6^s r \\
&+ 7.2383937069793\left(\alpha_0^s\right)^3\alpha_1^s\alpha_2^s r + 3.5455014290049\alpha_0^s\alpha_1^s\alpha_4^s r
\end{aligned}
$$

$$
\begin{aligned}
&+ 34.262761764317\alpha_0^s\alpha_1^s\alpha_2^s r^3 + 4.7242546387078\alpha_0^s\alpha_2^s\alpha_3^s r \\
&+ 60.649934470602\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r^2 \\
&- 19.591916445006\alpha_0^s\alpha_1^s\alpha_3^s r^2 - 11.224986566674\alpha_0^s\left(\alpha_1^s\right)^2\alpha_2^s r \\
&- 6.4968417518937\left(\alpha_0^s\right)^2\alpha_1^s\alpha_3^s r
\end{aligned}
$$

2.4 $n = 3$

For $p = 6$, the absolute cutoff is $W = 10$:

$$\alpha_3^t = \sum_{d=4}^{10} R_{3,d} + O(\mathrm{SR}^{11}).$$

Absolute order 4.

$$
\begin{aligned}
R_{3,4} = {} & 0.00146484375r^4 + 0.125\alpha_2^s r \\
&+ 0.125\left(\alpha_0^s\right)^3 r + 0.0234375\alpha_0^s r^3 \\
&+ 0.109375\left(\alpha_0^s\right)^2 r^2 - 0.0625\alpha_1^s r^2 \\
&- 0.375\alpha_0^s\alpha_1^s r
\end{aligned}
$$

Absolute order 5.

$$
\begin{aligned}
R_{3,5} = {} & -0.000762939453125r^5 + 0.125\alpha_3^s r \\
&+ 0.0419921875\alpha_1^s r^3 - 0.125\left(\alpha_0^s\right)^4 r \\
&- 0.375\left(\alpha_1^s\right)^2 r - 0.078125\alpha_2^s r^2 \\
&- 0.28125\left(\alpha_0^s\right)^3 r^2 - 0.01708984375\alpha_0^s r^4 \\
&- 0.1201171875\left(\alpha_0^s\right)^2 r^3 + 0.75\left(\alpha_0^s\right)^2\alpha_1^s r \\
&+ 0.4375\alpha_0^s\alpha_1^s r^2 - 0.5\alpha_0^s\alpha_2^s r
\end{aligned}
$$

Absolute order 6.

$$
\begin{aligned}
R_{3,6} = {} & 0.00069887861331449r^6 + 0.125\alpha_4^s r \\
&+ 0.52264594794504\left(\alpha_1^s\right)^2 r^2 + 0.10280837917801\left(\alpha_0^s\right)^5 r \\
&+ 0.52433019631832\left(\alpha_0^s\right)^3 r^3 + 0.017257388611671\alpha_0^s r^5 \\
&+ 0.07369519446916\alpha_2^s r^3 + 0.1495700174229\left(\alpha_0^s\right)^2 r^4 \\
&+ 0.61990642191755\left(\alpha_0^s\right)^4 r^2 - 0.042328393296221\alpha_1^s r^4 \\
&- 0.1038270947945\alpha_3^s r^2 + 1.6087005501362\alpha_0^s\left(\alpha_1^s\right)^2 r \\
&+ 1.0946586542461\left(\alpha_0^s\right)^2\alpha_2^s r + 0.71467513753404\alpha_0^s\alpha_2^s r^2 \\
&- 0.57385180946901\alpha_0^s\alpha_1^s r^3 - 1.9544879813691\left(\alpha_0^s\right)^2\alpha_1^s r^2 \\
&- 1.1168502750681\alpha_1^s\alpha_2^s r - 1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r \\
&- 0.58061675835603\alpha_0^s\alpha_3^s r
\end{aligned}
$$

Absolute order 7.

$$
\begin{aligned}
R_{3,7} = {} & -0.00066040799045703r^7 + 0.125\alpha_5^s r \\
&+ 0.044182253542082\alpha_1^s r^5 + 0.10547078563903\alpha_3^s r^3 \\
&+ 1.0840862194786\left(\alpha_1^s\right)^3 r - 0.12260923390637\alpha_4^s r^2 \\
&- 0.065244100954277\left(\alpha_0^s\right)^6 r - 1.8011431833565\left(\alpha_0^s\right)^4 r^3 \\
&- 1.1197437723225\left(\alpha_0^s\right)^5 r^2 - 0.076712092161027\alpha_2^s r^4 \\
&- 0.70760825653778\left(\alpha_1^s\right)^2 r^3 - 0.19445546235151\left(\alpha_0^s\right)^2 r^5
\end{aligned}
$$

$$
\begin{aligned}
&- 0.018648400200653\alpha_0^s r^6 - 0.81762161375293\left(\alpha_2^s\right)^2 r \\
&- 0.9077746502428\left(\alpha_0^s\right)^3 r^4 + 0.75724388321833\alpha_0^s\alpha_1^s r^4 \\
&+ 1.2633539166158\left(\alpha_0^s\right)^2\alpha_3^s r + 5.8694148411763\left(\alpha_0^s\right)^3\alpha_1^s r^2 \\
&+ 3.899202830862\left(\alpha_0^s\right)^2\alpha_1^s r^3 + 1.0138030827023\left(\alpha_0^s\right)^4\alpha_1^s r \\
&+ 1.7276736271957\alpha_1^s\alpha_2^s r^2 + 0.96375532208646\alpha_0^s\alpha_3^s r^2 \\
&- 1.0111179362879\alpha_0^s\alpha_2^s r^3 - 0.60587387125098\alpha_0^s\alpha_4^s r \\
&- 3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r \\
&- 4.6296729998974\alpha_0^s\left(\alpha_1^s\right)^2 r^2 - 3.2598096829233\left(\alpha_0^s\right)^2\alpha_2^s r^2 \\
&- 1.2984954850039\alpha_1^s\alpha_3^s r - 1.4700626232957\left(\alpha_0^s\right)^3\alpha_2^s r \\
&+ 4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r
\end{aligned}
$$

Absolute order 8.

$$
\begin{aligned}
R_{3,8} = {} & 0.00068532023746048 r^8 + 0.125\alpha_6^s r \\
&+ 4.3619461768873\left(\alpha_0^s\right)^4 r^4 + 0.038536574146577\left(\alpha_0^s\right)^7 r \\
&+ 5.2029254536162\left(\alpha_0^s\right)^5 r^3 + 0.021703788578011\alpha_0^s r^7 \\
&+ 1.545267118531\left(\alpha_0^s\right)^3 r^5 + 1.7512659597646\left(\alpha_0^s\right)^6 r^2 \\
&+ 1.010573413351\left(\alpha_1^s\right)^2 r^4 + 0.26453516473845\left(\alpha_0^s\right)^2 r^6 \\
&+ 0.13887279321988\alpha_4^s r^3 + 0.087120557839057\alpha_2^s r^5 \\
&+ 1.524715047639\left(\alpha_2^s\right)^2 r^2 - 3.8575245524135\left(\alpha_1^s\right)^3 r^2 \\
&- 0.05080989468861\alpha_1^s r^6 - 0.13952053443311\alpha_5^s r^2 \\
&- 0.12024763108045\alpha_3^s r^4 + 1.1971819419038\alpha_0^s\alpha_4^s r^2 \\
&+ 10.320494578314\left(\alpha_0^s\right)^3\alpha_2^s r^2 + 1.4593688359096\alpha_0^s\alpha_2^s r^4 \\
&+ 4.7937562503095\left(\alpha_1^s\right)^2\alpha_2^s r + 10.176804101055\alpha_0^s\left(\alpha_1^s\right)^2 r^3 \\
&+ 3.3523814346221\alpha_0^s\left(\alpha_2^s\right)^2 r + 7.272533332723\left(\alpha_0^s\right)^2\alpha_2^s r^3 \\
&+ 4.3030456716032\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r + 2.4584809206766\alpha_1^s\alpha_3^s r^2 \\
&+ 1.355409833533\left(\alpha_0^s\right)^2\alpha_4^s r \\
&+ 21.892191226939\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^2 \\
&+ 1.5128907607855\left(\alpha_0^s\right)^4\alpha_2^s r - 1.7733901443961\left(\alpha_0^s\right)^3\alpha_3^s r \\
&- 1.0496651482817\alpha_0^s\alpha_1^s r^5 - 4.6393166739312\left(\alpha_0^s\right)^2\alpha_3^s r^2 \\
&- 0.61616427546487\alpha_0^s\alpha_5^s r - 13.841809923869\left(\alpha_0^s\right)^4\alpha_1^s r^2 \\
&- 4.4309930101446\alpha_0^s\left(\alpha_1^s\right)^3 r - 18.743138308883\left(\alpha_0^s\right)^3\alpha_1^s r^3 \\
&- 2.6475317332077\alpha_1^s\alpha_2^s r^3 - 1.3805669230387\alpha_1^s\alpha_4^s r \\
&- 1.5289756803498\alpha_0^s\alpha_3^s r^3 - 0.83496561933704\left(\alpha_0^s\right)^5\alpha_1^s r \\
&- 7.2513766025917\left(\alpha_0^s\right)^2\alpha_1^s r^4 - 1.9358793917918\alpha_2^s\alpha_3^s r \\
&+ 5.4470368262232\alpha_0^s\alpha_1^s\alpha_3^s r - 16.335385878746\alpha_0^s\alpha_1^s\alpha_2^s r^2 \\
&- 9.3326987690427\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r
\end{aligned}
$$

Absolute order 9.

$$
\begin{aligned}
R_{3,9} = {} & -0.00077806890581001 r^9 + 0.125\alpha_7^s r \\
&+ 0.06358704244356\alpha_1^s r^7 + 9.5157792367703\left(\alpha_1^s\right)^3 r^3 \\
&+ 0.15020868322051\alpha_3^s r^5 + 0.17525234851113\alpha_5^s r^3 \\
&- 2.6786068018394\left(\alpha_0^s\right)^3 r^6 - 2.4869134686556\left(\alpha_1^s\right)^4 r
\end{aligned}
$$

$$
\begin{aligned}
&- 1.1877542095598\left(\alpha_3^s\right)^2 r - 0.15572253060722\alpha_6^s r^2 \\
&- 1.5189127765673\left(\alpha_1^s\right)^2 r^5 - 17.933550190782\left(\alpha_0^s\right)^5 r^4 \\
&- 0.020902477635746\left(\alpha_0^s\right)^8 r - 13.339047064301\left(\alpha_0^s\right)^6 r^3 \\
&- 9.8616620914994\left(\alpha_0^s\right)^4 r^5 - 2.4713378521286\left(\alpha_0^s\right)^7 r^2 \\
&- 2.6843690618889\left(\alpha_2^s\right)^2 r^3 - 0.17465185173482\alpha_4^s r^4 \\
&- 0.38149897445917\left(\alpha_0^s\right)^2 r^7 - 0.10897896751649\alpha_2^s r^6 \\
&- 0.02733787086542\alpha_0^s r^8 + 9.6482197902839\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^3 r \\
&+ 1.5394198894557\alpha_0^s\alpha_1^s r^6 + 2.4542491246068\alpha_0^s\alpha_3^s r^4 \\
&+ 11.804717477222\left(\alpha_0^s\right)^2\alpha_3^s r^3 + 72.21470484842\left(\alpha_0^s\right)^4\alpha_1^s r^3 \\
&+ 1.4028073758501\left(\alpha_0^s\right)^2\alpha_5^s r + 3.2619032889355\alpha_1^s\alpha_4^s r^2 \\
&+ 13.367356356375\left(\alpha_0^s\right)^2\alpha_1^s r^5 + 1.9057413958323\left(\alpha_0^s\right)^4\alpha_3^s r \\
&+ 15.732895949371\left(\alpha_0^s\right)^3\alpha_3^s r^2 + 4.1955896290722\alpha_1^s\alpha_2^s r^4 \\
&+ 4.7106441588735\alpha_2^s\alpha_3^s r^2 + 0.60426087709863\left(\alpha_0^s\right)^6\alpha_1^s r \\
&+ 38.920069106575\alpha_0^s\left(\alpha_1^s\right)^3 r^2 + 1.4322862272868\alpha_0^s\alpha_5^s r^2 \\
&+ 6.0326847252929\left(\alpha_1^s\right)^2\alpha_3^s r + 49.631939118393\left(\alpha_0^s\right)^3\alpha_1^s r^4 \\
&+ 27.575781873901\left(\alpha_0^s\right)^5\alpha_1^s r^2 + 7.4740282968277\alpha_1^s\left(\alpha_2^s\right)^2 r \\
&- 0.62078024485779\alpha_0^s\alpha_6^s r - 2.2151677365237\alpha_0^s\alpha_2^s r^5 \\
&- 4.6768197754514\left(\alpha_0^s\right)^4\left(\alpha_1^s\right)^2 r \\
&- 78.451840902511\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^3 \\
&- 20.710877845769\alpha_0^s\left(\alpha_1^s\right)^2 r^4 - 1.4263320561407\alpha_1^s\alpha_5^s r \\
&- 2.1183060208428\alpha_2^s\alpha_4^s r - 2.1443384104247\alpha_0^s\alpha_4^s r^3 \\
&- 4.3219288525125\alpha_1^s\alpha_3^s r^3 - 73.243320022063\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r^2 \\
&- 37.486758705034\left(\alpha_0^s\right)^3\alpha_2^s r^3 - 1.3089551224142\left(\alpha_0^s\right)^5\alpha_2^s r \\
&- 22.035682750882\left(\alpha_1^s\right)^2\alpha_2^s r^2 - 6.1322897063433\left(\alpha_0^s\right)^2\alpha_4^s r^2 \\
&- 26.018233363777\left(\alpha_0^s\right)^4\alpha_2^s r^2 \\
&- 7.2650185519848\left(\alpha_0^s\right)^2\left(\alpha_2^s\right)^2 r \\
&- 14.97896466591\left(\alpha_0^s\right)^2\alpha_2^s r^4 - 15.608461554937\alpha_0^s\left(\alpha_2^s\right)^2 r^2 \\
&- 1.9617569380588\left(\alpha_0^s\right)^3\alpha_4^s r + 13.488476001963\left(\alpha_0^s\right)^3\alpha_1^s\alpha_2^s r \\
&+ 40.919197164152\alpha_0^s\alpha_1^s\alpha_2^s r^3 + 6.0176732098989\alpha_0^s\alpha_1^s\alpha_4^s r \\
&+ 8.3704075852001\alpha_0^s\alpha_2^s\alpha_3^s r + 82.879921620928\left(\alpha_0^s\right)^2\alpha_1^s\alpha_2^s r^2 \\
&- 20.840808859589\alpha_0^s\left(\alpha_1^s\right)^2\alpha_2^s r \\
&- 11.747481990311\left(\alpha_0^s\right)^2\alpha_1^s\alpha_3^s r - 25.047186783622\alpha_0^s\alpha_1^s\alpha_3^s r^2
\end{aligned}
$$

Absolute order 10.

$$
\begin{aligned}
R_{3,10} = {} & 0.00096292509101481 r^{10} + 0.125\alpha_8^s r \\
&+ 4.0935689153387\left(\alpha_2^s\right)^3 r + 4.8079900360626\left(\alpha_0^s\right)^3 r^7 \\
&+ 28.170001262946\left(\alpha_1^s\right)^4 r^2 + 0.037210311634069\alpha_0^s r^9 \\
&+ 0.011049369381434\left(\alpha_0^s\right)^9 r + 0.58554447056849\left(\alpha_0^s\right)^2 r^8 \\
&+ 4.7559702041192\left(\alpha_2^s\right)^2 r^4 + 2.4184043629563\left(\alpha_1^s\right)^2 r^6 \\
&+ 21.905690736705\left(\alpha_0^s\right)^4 r^6 + 0.2149437032619\alpha_6^s r^3 \\
&+ 0.2392938565592\alpha_4^s r^5 + 4.0041747332972\left(\alpha_3^s\right)^2 r^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 66.175974906622\left(\alpha_0^s\right)^6 r^4 + 0.14837237070063\alpha_2^s r^7 \\
&+ 3.2409184629156\left(\alpha_0^s\right)^8 r^2 + 54.258044816224\left(\alpha_0^s\right)^5 r^5 \\
&+ 31.453535336483\left(\alpha_0^s\right)^7 r^3 - 21.384166308953\left(\alpha_1^s\right)^3 r^4 \\
&- 0.17161851595073\alpha_7^s r^2 - 0.086355279888405\alpha_1^s r^8 \\
&- 0.20536304853596\alpha_3^s r^6 - 0.24166580521179\alpha_5^s r^4 \\
&+ 434.7513127713\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^2 r^3 \\
&+ 110.69159557962\left(\alpha_0^s\right)^3 \alpha_2^s r^4 + 4.1719758151042\alpha_1^s \alpha_5^s r^2 \\
&+ 6.8286791143162\alpha_2^s \alpha_4^s r^2 + 30.400776542534\left(\alpha_0^s\right)^2 \alpha_2^s r^5 \\
&+ 85.885091704801\left(\alpha_0^s\right)^2\left(\alpha_2^s\right)^2 r^2 \\
&+ 1.4276305967031\left(\alpha_0^s\right)^2 \alpha_6^s r + 11.578560735411\alpha_0^s\left(\alpha_1^s\right)^4 r \\
&+ 6.9015615546597\left(\alpha_1^s\right)^2 \alpha_4^s r + 55.820160715493\left(\alpha_0^s\right)^5 \alpha_2^s r^2 \\
&+ 7.6567059242057\alpha_1^s \alpha_3^s r^4 + 1.019898851088\left(\alpha_0^s\right)^6 \alpha_2^s r \\
&+ 2.1948156748004\left(\alpha_0^s\right)^4 \alpha_4^s r \\
&+ 4.3984922749124\left(\alpha_0^s\right)^5\left(\alpha_1^s\right)^2 r \\
&+ 17.763912905263\left(\alpha_0^s\right)^2 \alpha_4^s r^3 + 156.08530151767\left(\alpha_0^s\right)^4 \alpha_2^s r^3 \\
&+ 41.68950741035\alpha_0^s\left(\alpha_1^s\right)^2 r^5 + 5.4217853596071\alpha_0^s\left(\alpha_3^s\right)^2 r \\
&+ 22.272127960388\left(\alpha_0^s\right)^3 \alpha_4^s r^2 \\
&+ 11.198753592781\left(\alpha_0^s\right)^3\left(\alpha_2^s\right)^2 r \\
&+ 45.032565111185\alpha_0^s\left(\alpha_2^s\right)^2 r^3 \\
&+ 195.7067448819\left(\alpha_0^s\right)^4\left(\alpha_1^s\right)^2 r^2 + 1.6750834659867\alpha_0^s \alpha_6^s r^2 \\
&+ 3.5512161774335\alpha_0^s \alpha_2^s r^6 + 229.41201628325\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r^4 \\
&+ 3.819334211934\alpha_0^s \alpha_4^s r^4 + 62.592685527942\left(\alpha_1^s\right)^2 \alpha_2^s r^3 \\
&- 2.2290137046492\alpha_2^s \alpha_5^s r - 16.524894016307\left(\alpha_1^s\right)^3 \alpha_2^s r \\
&- 2.3999070134346\alpha_0^s \alpha_1^s r^7 - 42.688809375869\left(\alpha_0^s\right)^4 \alpha_3^s r^2 \\
&- 46.092785981788\alpha_1^s\left(\alpha_2^s\right)^2 r^2 - 4.1030991267942\alpha_0^s \alpha_3^s r^5 \\
&- 158.47402671634\alpha_0^s\left(\alpha_1^s\right)^3 r^3 - 25.140121908756\left(\alpha_0^s\right)^2 \alpha_1^s r^6 \\
&- 209.96248448079\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^3 r^2 \\
&- 27.126881179596\left(\alpha_0^s\right)^2 \alpha_3^s r^4 - 6.9521485719624\alpha_1^s \alpha_2^s r^5 \\
&- 239.34218243045\left(\alpha_0^s\right)^5 \alpha_1^s r^3 \\
&- 15.045237010191\left(\alpha_0^s\right)^3\left(\alpha_1^s\right)^3 r \\
&- 65.638645312796\left(\alpha_0^s\right)^3 \alpha_3^s r^3 - 2.0766487599458\left(\alpha_0^s\right)^3 \alpha_5^s r \\
&- 274.89784934706\left(\alpha_0^s\right)^4 \alpha_1^s r^4 - 36.854872353998\left(\alpha_1^s\right)^2 \alpha_3^s r^2 \\
&- 9.6012366793424\alpha_2^s \alpha_3^s r^3 - 0.62294812760585\alpha_0^s \alpha_7^s r \\
&- 2.6718595907698\alpha_3^s \alpha_4^s r - 2.8743200144418\alpha_0^s \alpha_5^s r^3 \\
&- 1.4501636832838\alpha_1^s \alpha_6^s r - 48.655464326521\left(\alpha_0^s\right)^6 \alpha_1^s r^2 \\
&- 0.40820845613314\left(\alpha_0^s\right)^7 \alpha_1^s r - 6.5595242162778\alpha_1^s \alpha_4^s r^3 \\
&- 1.7479347339582\left(\alpha_0^s\right)^5 \alpha_3^s r - 122.46906991114\left(\alpha_0^s\right)^3 \alpha_1^s r^5 \\
&- 7.7816176189195\left(\alpha_0^s\right)^2 \alpha_5^s r^2 + 72.231907558472\alpha_0^s \alpha_1^s \alpha_3^s r^3 \\
&+ 9.5465542166 84\alpha_0^s \alpha_2^s \alpha_4^s r \\
&+ 48.320543497115\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 \alpha_2^s r \\
&+ 137.3529810443\left(\alpha_0^s\right)^2 \alpha_1^s \alpha_3^s r^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 17.92663299468 \left(\alpha_0^s\right)^3 \alpha_1^s \alpha_3^s r + 19.623512720695 \alpha_1^s \alpha_2^s \alpha_3^s r \\
&+ 6.3570819228773 \alpha_0^s \alpha_1^s \alpha_5^s r + 240.96375402815 \alpha_0^s \left(\alpha_1^s\right)^2 \alpha_2^s r^2 \\
&- 92.980237691295 \alpha_0^s \alpha_1^s \alpha_2^s r^4 - 34.464261820289 \alpha_0^s \alpha_1^s \left(\alpha_2^s\right)^2 r \\
&- 35.80149408137 \alpha_0^s \alpha_1^s \alpha_4^s r^2 - 19.141816162204 \left(\alpha_0^s\right)^2 \alpha_2^s \alpha_3^s r \\
&- 299.4185696385 \left(\alpha_0^s\right)^3 \alpha_1^s \alpha_2^s r^2 \\
&- 341.62703102315 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_2^s r^3 \\
&- 52.4866485157 \alpha_0^s \alpha_2^s \alpha_3^s r^2 - 13.483242053444 \left(\alpha_0^s\right)^2 \alpha_1^s \alpha_4^s r \\
&- 27.574730196116 \alpha_0^s \left(\alpha_1^s\right)^2 \alpha_3^s r \\
&- 15.698042993856 \left(\alpha_0^s\right)^4 \alpha_1^s \alpha_2^s r
\end{aligned}
$$